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Influence of Recovery Time After Warm-up on Vertical Jump Performance in Trained Prepubertal and Postpubertal Male Athletes

Antoine Ferrari,¹ Georges Baquet,¹ Anthony J. Blazeovich,² and Sébastien Ratel³

¹Univ. Lille, Univ. Artois, Univ. Littoral Côte d'Opale, ULR 7369-URePSSS—Unité de Recherche Pluridisciplinaire Sport Santé Société, Lille, France; ²Centre for Human Performance, School of Medical and Health Sciences, Edith Cowan University, Joondalup, Australia; and ³Clermont-Auvergne University, AME2P, Clermont-Ferrand, France

Abstract

Ferrari, A, Baquet, G, Blazeovich, AJ, and Ratel, S. Influence of recovery time after warm-up on vertical jump performance in trained prepubertal and postpubertal male athletes. *J Strength Cond Res* XX(X): 000–000, 2023—The aim of this study was to examine the concurrent effects of maturity status and recovery delay after a standardized dynamic warm-up on vertical jump performance. Thirteen prepubertal boys (9.4 ± 1.1 years) and 12 postpubertal boys (16.6 ± 0.8 years) were tested on squat jump (SJ) and drop jump from 30 cm (DJ₃₀) before and after a standardized, 8-minute, dynamic warm-up, including low-intensity, nonspecific dynamic exercises progressing to maximal, specific exercises, over 6 separate occasions. In each session, subjects jumped at 0, 1.5, 3, 5, 7, or 10 minutes after warm-up in a randomized order. Measurements included SJ height, DJ₃₀ height, ground contact time (GCT), and reactive strength index (RSI) from DJ₃₀. The results revealed no significant recovery time $\times$ maturity group interaction effects for SJ height ($p = 0.36$), DJ₃₀ height ($p = 0.45$), GCT ($p = 0.75$), or RSI ($p = 0.09$), meaning that maturity status did not have a significant effect on changes in vertical jump performance after the warm-up. However, there were significant time effects for SJ height, DJ₃₀ height, and RSI ($p < 0.001$ for all), with DJ₃₀ height and RSI increasing significantly by 16.9% at 1.5 minutes ($p < 0.001$) and SJ height increasing significantly by 12.0% until 3 minutes after the warm-up ($p < 0.001$). To conclude, the standardized dynamic warm-up had beneficial effects on vertical jump performance within the first 3 minutes after completion of the intervention. However, vertical jump performance after the warm-up was not dependent on the children's maturity status.

Key Words: maturation, recovery delay

Introduction

The practice of warm-up before physical exertion has been widespread both for the improvement of athletic performance and prevention of potential sports injuries. In the adult population, the benefits of warm-up on short-term high-intensity exercise performance have been commonly attributed to several physiological mechanisms underpinning postactivation potentiation (PAP) of muscle contractile properties (e.g., myosin regulatory light chain phosphorylation) and postactivation performance enhancement (PAPE) effects driven by increased muscle temperature, muscle and muscle fiber water content, and muscle activation and coordination (12).

However, in the pediatric population, few studies have examined the effects of warm-up on sports performance (2,14–16,24,28,30), and although most acute dynamic warm-up protocols showed beneficial effects on short-term high-intensity exercise performance during childhood and adolescence, many of them were not sufficiently well designed to precisely evaluate the acute effects of dynamic warm-up protocols on performance in athletic youngsters. For instance, some studies compared different warm-up protocols using static stretching, dynamic stretching, or dynamic exercises without having determined baseline pre-warm-up performance, thus biasing evaluation of acute effects of

each warm-up protocol and their comparison on performance in children and adolescents (14,15,24,28). Furthermore, some studies pooled prepubertal boys and girls to compare different warm-up protocols on sports performance (14), whereas the PAP effect (increase in twitch force) on the plantar flexor muscles was found to be greater in prepubertal boys than in prepubertal girls (26). In addition, most studies evaluated the acute effect of warm-up at only one time point after the completion of warm-up (i.e., between 1 and 2 minutes in children and teenage athletes (14,15) or between 8 and 10 minutes in teenage athletes) (30), whereas recovery time on dynamic warm-up effects was found to be optimal around the sixth minute postwarm-up in teenage athletes (16). Thus, in the light of these methodological considerations, it is possible to believe that the acute effects of warm-up on short-term high-intensity exercise performance have been overestimated or underestimated during childhood and adolescence depending on undetermined baseline pre-warm-up performance, sex, or recovery delay following dynamic warm-up protocols.

Nonetheless, other factors such as children's maturity status could also account for the evaluation of dynamic warm-up effects because the warm-up effect on squat jump (SJ) performance and peak rate of force development was found to be age and maturity dependent (2). Indeed, Arabatzi et al. (2) showed that SJ height improved in response to 3 maximal isometric half-squats performed 15 minutes after an 8-minute general warm-up in men but not in prepubertal boys or pubertal male adolescents, whereas its

Address correspondence to Dr. Sébastien Ratel, Sebastien.RATEL@uca.fr.

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effect on the peak rate of force development during the jump was significant in men and pubertal male adolescents but not in prepubertal boys. In addition, recovery delay after warm-up has never been analyzed with respect to maturity status of children, whereas dynamic warm-up–derived fatigue could be greater in postpubertal than in prepubertal children (27), and a greater fatigue could overcome the physiological mechanisms underlying the PAP effect (29). However, direct scientific evidence for the concurrent effects of time recovery and maturity status on short-term high-intensity exercise performance has not been specifically sought. More importantly, the relative effects of longer, comprehensive, warm-up programs that might be more commonly used in sports practice between prepubertal and postpubertal children are unknown. The examination of these potential effects in young athletes is important because warm-up programs, including low-intensity, nonspecific exercises progressing to maximal, specific exercises, are often used in sport settings for increasing jumping, sprinting, and strength abilities and for optimizing athletic performance.

The purpose of this study was to examine the concurrent effects of maturity status and recovery time after a standardized dynamic warm-up usually done in a sporting setting on short-term high-intensity exercise performance. It was hypothesized that a standardized dynamic warm-up period, including low-intensity, nonspecific exercises progressing to maximal specific exercises, would have greater beneficial effects on short-term high-intensity exercise performance (i.e., vertical jump performance) in postpubertal than in prepubertal male youngsters. These benefits in postpubertal athletes could be greater during the SJ than during drop jump (DJ) because changes in muscle power during the SJ should theoretically result from increases in muscle temperature (8) whereas the DJ performance, which requires a stretch-shortening cycle (SSC) and coordination ability, could be too complex to be influenced by a simple standardized dynamic warm-up.

Methods

AU4 Experimental Approach to the Problem

In this study, we wanted to examine the effects of recovery time from a standardized dynamic warm-up on selected vertical jump tests SJ and drop jump from 30 cm (DJ₃₀) in relation to the maturity status of children. Following baseline testing on SJ and DJ₃₀, 2 trained groups of prepubertal and postpubertal children participated in an 8-minute warm-up protocol consisting of dynamic exercises (Table 1). Subsequently, SJ and DJ₃₀ performances were assessed at 0, 1.5, 3, 5, 7 or 10 minutes after warm-up. In total, each subject completed 6 experimental sessions preceded by 2 introductory sessions on nonconsecutive days, which were separated by at least 48 hours. The 6 experimental sessions

were done in a randomized order; testing across days prevented between-test performance enhancement or interference.

Subjects

Thirteen boys (9.4 ± 1.1 years) and 12 male adolescents (16.6 ± 0.8 years) volunteered to participate in this study. The physical characteristics of subjects are described in the Table 1.

The sample size estimation was determined for a 2-sided type I error at 5% and a statistical power greater than 80%, on the basis of previously published data in the male population (2). To be included, boys had to train for ≥ 3 hours per week in addition to physical education lessons at school (2 hours per week) but not in a specific sport or physical conditioning setting. They also had to be free of any medical contraindication to sporting activity. They were prepubertal based on the assessment of their somatic maturity (see below). Boys were recruited from primary schools or local sports clubs. The male adolescents were postpubertal. They practiced rugby 3 times per week, resistance training 2 times a week for at least one year, and 2 hours of physical education at school. This study was approved by a local institutional ethics review board (Ethics committee for research, Lille University, n°2021-507-S94). The study was conducted in conformity with the policy statement regarding the use of human subjects by the Declaration of Helsinki. All experimental procedures were clearly explained to the subjects, who then gave written assent before the commencement of testing. Written consent with signatures was also obtained from parents or guardians before boys were accepted in the study.

Procedures

AU5

Before data collection, all volunteers participated in 2 introductory sessions during which they followed all warm-up procedures and practiced vertical jumps (SJ and DJ₃₀). The introductory period was designed to reduce the influence of learning effects on SJ and DJ₃₀ performances and to ensure proper execution of the warm-up exercises. Following baseline testing on the vertical jumps, volunteers then participated in an 8-minute warm-up protocol that consisted of dynamic exercises (see below for further details). Subsequently, SJ and DJ₃₀ performances were assessed at 0, 1.5, 3, 5, 7 or 10 minutes after warm-up on nonconsecutive days separated by at least 48 hours (i.e., over 6 separate experimental sessions). Squat jump performance was systematically assessed before DJ₃₀ performance to account for any small potentiating effect of the first jump on the second at each post-warm-up time point. All the tests were performed indoor at the same time of day with a temperature between 22° C and 27° C. During each session, subjects warmed up in groups of 4–5 under close supervision by the same research assistant who demonstrated the proper technique for SJ and DJ₃₀ movements. Volunteers continued to perform the same type of training as usual throughout the protocol but refrained from participating in any vigorous physical activity for 24 hours before any testing session. Each subject completed all study procedures within 21 days. Testing at each time interval was complete within 30 seconds. During the recovery period, the subjects had to remain seated to standardize the experimental conditions over the 6 separate sessions. The same research assistant tested the subjects on SJ and DJ₃₀ throughout the study period. Squat jump and DJ₃₀ tests were chosen because the former is largely considered to be a measure of concentric muscle performance, whereas the latter

Table 1
Subjects' physical characteristics (mean $\pm$ SD).

	Prepubertal boys	Postpubertal boys
Age (y)	9.4 ± 1.1	$16.6 \pm 0.8^*$
Body mass (kg)	33.7 ± 4.7	$88.2 \pm 15.9^*$
Height (cm)	140.5 ± 6.7	$181 \pm 6.4^*$
Sitting height (cm)	71.7 ± 3.1	$154.8 \pm 3.9^*$
Leg length (cm)	68.8 ± 4.8	$86.5 \pm 3.5^*$
Maturity offset (y)	-3.6 ± 0.7	$3.0 \pm 0.8^*$

*Significantly different between prepubertal and postpubertal boys from $p < 0.001$.

involves both eccentric and concentric phases and requires a complex movement strategy. In addition, DJ is typically examined in research studies because it is frequently used in lower-body plyometric training programs (10).

Anthropometric Characteristics and Maturity Status Assessment.

AUG

Body mass (BM) was measured using a digital weight scale (TANITA, BC-545N, Japan), and standing height was assessed using a portable stadiometer with the subjects barefoot (TANITA, HR001). Sitting height was measured with the stadiometer whereas the subjects sat on the floor with their back against a wall. Maturity offset (MO in years) was determined to assess somatic maturity (i.e., years to [from] age at peak height velocity [APHV]) using chronological age, BM, standing height, and sitting height. Its calculation was done based on sex-specific regression equations according to the method proposed by Mirwald et al. (23). Subjects were considered as prepubertal or postpubertal based on their MO (pre-pubertal: < -1 year from PHV and postpubertal: > +1 year from PHV). Maturity offset was found to be estimated within an error of ± 1 year, which is sufficient for adolescence to be assigned a maturational classification (23).

Warm-up Program. The 8-minute, standardized, dynamic warm-up was consistent with that of Petrakis et al. (28) and consisted of 3-minute running, 4-minute dynamic movements, and 1-minute vertical jumps (4 in total, described below) at maximum intensity. The running intensity was determined from the Children's Effort Rating Table (CERT), which is validated to prescribe the intensity of various forms of physical activity in children (37). The subjects were asked to run at a level-2 intensity ("very easy") for 2 minutes, then at a level-4 intensity for 1 minute ("just feeling a strain"). During the following 4 minutes, the children performed dynamic movements over a 10-m distance; they had to walk back to the starting point after each exercise (see Table 2 for further details). During this phase, the speed of movement and plyometric intensity of the exercises were progressively increased. Dynamic exercises were preferred to static stretching exercises to eliminate any potential adverse effect on vertical jump performance (16). Finally, the children performed 2 SJ and then 2 DJ₃₀ at maximum intensity over 1 minute (i.e., 15 seconds between attempts).

T2

Table 2
Dynamic warm-up exercises.

1.	Walking lunge. Large front lunge with alternate right and left legs (back knee does not touch the ground)
2.	Knee to chest walks. With each step, raise the knee up to the chest by pulling it with both arms
3.	High knees. Move forward with knee lift, arm swing, and frequency
4.	Heel flicks. Move forward with heel back, arm swing, and frequency
5.	Straight leg running. Move forward with straight leg, arm swing, and frequency
6.	Side shuffle (left and right). Move laterally quickly without crossing feet
7.	Carioca drills (left and right). Move laterally quickly, with alternating crossing the back leg in front and behind the other leg
8.	Bounding. Move forward with largest steps as possible and with minimum ground contact time
9.	Front leg swing. Swing one leg forward and backward with maximum amplitude (10 repetitions per side)
10.	Lateral leg swing. Swing one leg side to side with maximum amplitude (10 repetitions per side)

Testing. The jumping tests were evaluated using an optical measurement system (OptoJump; Microgate, Bolzano, Italy). This system has previously been validated for the collection and analysis of jump performance (4).

Squat Jump From a standing position, the subjects flexed their knees to lower their center of mass (i.e., to a squatting position) to a comfortable position with knee angle between 80° and 100° (self-selected). The hands were placed on the hips with the elbows bowed outward. This position was held for 3 seconds before a maximal vertical jump was performed without countermovement. Jumps were repeated if the researcher could clearly visually detect a countermovement before the upward (jump) phase. To ensure that the push-off distance was the same at each SJ for each subject, the starting position was measured during the introductory sessions and monitored over the 6 experimental sessions. To measure the starting position, we checked the knee angle with a manual goniometer before each attempt.

Drop Jump from 30 cm A 30-cm box was placed in front of the OptoJump system. From a standing position close to the front edge of the box, the children extended one foot out in front of themselves to allow their body to move forward over the edge of the box before letting their body fall vertically without pushing upward or outward, jumping, or flexing their support leg from the box. When done properly, the subjects landed close to the box (heel < 25 cm from the front of the box). Upon landing with both feet, they immediately jumped as high as possible. In their native French language, the instruction to the children was to "push as strong as possible and reduce the ground contact time." They were asked to keep their legs straight and feet plantar flexed upon landing. The height of 30 cm was selected because it has been found to be in the optimal range to optimize DJ performance in children and adolescents (10).

Measurements and Calculations.

Squat Jump Performance In accordance with the fundamental laws of dynamics, the SJ height (H_{SJ} in cm) was determined from aerial time (t_{SJ} in s) using the following equation:

$$H_{SJ} = 1/8 \cdot g \cdot t_{SJ}^2, \quad (1)$$

where t_{SJ} was measured with the OptoJump system and g corresponds to the acceleration because of gravity ($g = 9.81 \text{ m} \cdot \text{s}^{-2}$).

Drop Jump from 30-cm Performance In addition to DJ₃₀ height, ground contact time (GCT) was also retained from the drop jumps and the reactive strength index (RSI), considered as a measurement of "explosive strength" (33), was calculated by dividing DJ₃₀ height by GCT during the drop jumps.

Statistical Analyses

Experimental data are expressed as mean $\pm$ SD. Coefficient of variations (CV), intraclass correlations (ICC), 95% confidence intervals (CIs), and typical errors (TE) (18) were calculated for SJ (height) and DJ (height, GCT, RSI). Normality of the data was checked with the Shapiro-Wilk test. The effects of recovery time, maturity status, and their interaction on the dependent variables (SJ and DJ₃₀ heights, GCT, RSI) were analyzed using a 2-way analysis of variance with repeated measures. If the F value was

significant, Bonferroni's post hoc tests were applied. The effect size and statistical power have also been reported when significant main or interaction effects were detected. The effect size was assessed using the partial eta-squared (η^2) and ranked as follows: approximately 0.01 = small effect, approximately 0.06 = moderate effect, and ≥ 0.14 = large effect (13). In all analyses, the level of significance was set at $p < 0.05$. Data were analyzed with Prism 9.3 (Graphpad Software Inc., La Jolla, CA).

Results

Reliability Outcomes

Intertrial test-retest reliability for pre-warm-up outcomes over the 6 experimental sessions are displayed in Table 3.

T3 Intraclass correlation coefficient, TE, and CV are criteria for reliability. Reliability is generally considered high when ICC is above 0.90 (35) and good when ICC is above 0.75 (13). In prepubertal boys, reliability was high for RSI (ICC = 0.91), good for SJ height (ICC = 0.79), and DJ₃₀ height (ICC = 0.78). In postpubertal boys, reliability was high for SJ height (ICC = 0.92), good for DJ₃₀ height (ICC = 0.75), and moderate for RSI (ICC = 0.72). For GCT, reliability was moderate in prepubertal boys (ICC = 0.60) and low in postpubertal boys (ICC = 0.24). Typical error was low for all tests, which indicates a good reliability as a smaller TE indicates a higher reliability (3). Coefficient of variations are generally considered acceptable when $<10\%$ (3). Coefficient of variations were acceptable for all outcomes except DJ₃₀ height and GCT in prepubertal boys.

Effects of Maturity and Recovery Time

The recovery time $\times$ maturity group interaction effects are displayed in Table 4. Analysis of variance revealed no significant recovery time $\times$ maturity group interaction effects for SJ height ($p = 0.36$), DJ₃₀ height ($p = 0.45$), GCT ($p = 0.75$), and RSI ($p = 0.09$).

Data for SJ height, DJ₃₀ height, and GCT before warm-up and across the different post-warm-up time points are displayed in Table 5. Vertical jump outcomes were significantly higher in **T5** postpubertal compared with prepubertal boys ($p < 0.001$) except for GCT.

Bonferroni's multicomparisons test outcomes for recovery time effect are presented in Table 6. After warm-up, DJ₃₀ height and **T6** RSI increased significantly until 1.5 minutes, whereas SJ height increased significantly until 3 minutes. The time courses of SJ height, DJ₃₀ height, and RSI across the different post-warm-up time points are displayed in Figure 1.

F1

Discussion

This study examined the influence of recovery time after warm-up on vertical jump performance in prepubertal and postpubertal boys. No significant recovery time $\times$ maturity group interaction effects were detected on any parameter measured during vertical jumps, suggesting that the maturity status of boys did not influence changes in vertical jump performance after completion of the warm-up. However, the results of this study showed significant main effects for recovery time, with an increase of SJ height being detected until 3 minutes after warm-up ($+12.0\%$) and increases in DJ₃₀ height and RSI until 1.5 minutes after warm-up ($+16.9\%$).

It is well acknowledged that force production capacity is greater in postpubertal than in prepubertal children (34). This

Table 3 Intertrial test-retest reliability for pre-warm-up outcomes over the 6 experimental sessions.*†

	Session 1	Session 2	Session 3	Session 4	Session 5	Session 6	TE (95% CI)	ICC (95% CI)	CV (%) (95% CI)
Prepubertal boys									
SJ height (m)	0.164 $\pm$ 0.024	0.169 $\pm$ 0.038	0.165 $\pm$ 0.025	0.164 $\pm$ 0.031	0.168 $\pm$ 0.028	0.164 $\pm$ 0.024	0.012 (0.009–0.015)	0.86 (0.73–0.95)	7.5 (6.2–11.1)
DJ ₃₀ height (m)	0.170 $\pm$ 0.036	0.172 $\pm$ 0.042	0.175 $\pm$ 0.033	0.170 $\pm$ 0.029	0.172 $\pm$ 0.03	0.170 $\pm$ 0.036	0.017 (0.014–0.022)	0.78 (0.59–0.91)	10.8 (8.9–13.8)
GCT (s)	0.236 $\pm$ 0.034	0.242 $\pm$ 0.028	0.229 $\pm$ 0.028	0.230 $\pm$ 0.039	0.228 $\pm$ 0.04	0.236 $\pm$ 0.034	0.025 (0.021–0.031)	0.60 (0.36–0.82)	11.0 (9.1–14.2)
RSI	0.740 $\pm$ 0.219	0.721 $\pm$ 0.195	0.788 $\pm$ 0.226	0.727 $\pm$ 0.254	0.784 $\pm$ 0.225	0.740 $\pm$ 0.219	0.070 (0.059–0.089)	0.91 (0.82–0.97)	7.8 (7.5–11.6)
Postpubertal boys									
SJ height (m)	0.330 $\pm$ 0.048	0.324 $\pm$ 0.049	0.323 $\pm$ 0.057	0.323 $\pm$ 0.046	0.325 $\pm$ 0.042	0.330 $\pm$ 0.052	0.016 (0.013–0.020)	0.92 (0.82–0.97)	4.9 (3.9–7.0)
DJ ₃₀ height (m)	0.381 $\pm$ 0.046	0.364 $\pm$ 0.047	0.370 $\pm$ 0.055	0.369 $\pm$ 0.046	0.365 $\pm$ 0.061	0.365 $\pm$ 0.062	0.028 (0.023–0.036)	0.75 (0.54–0.91)	6.7 (5.3–9.5)
GCT (s)	0.243 $\pm$ 0.018	0.240 $\pm$ 0.021	0.238 $\pm$ 0.017	0.242 $\pm$ 0.019	0.241 $\pm$ 0.017	0.240 $\pm$ 0.024	0.017 (0.014–0.022)	0.24 (0.02–0.58)	7.3 (5.8–10.4)
RSI	1.574 $\pm$ 0.219	1.524 $\pm$ 0.233	1.560 $\pm$ 0.213	1.534 $\pm$ 0.203	1.514 $\pm$ 0.213	1.529 $\pm$ 0.280	0.122 (0.102–0.157)	0.74 (0.53–0.90)	9.1 (7.2–13.0)

*SJ = squat jump; DJ₃₀ = drop jump from 30 cm; GCT = ground contact time; RSI = reactive strength index; TE = typical error; ICC = intraclass coefficient correlation; CV = coefficient of variation; CI = confidence interval.
†Mean $\pm$ SD.

Table 4
Two-way ANOVA results for the different vertical jump outcomes.*

	SS	DF	MS	F (DFn, DFd)	p	η^2	Statistical power
DJ ₃₀ height							
Time × maturity	0.004040	6	0.0006733	$F(6, 138) = 0.97$	0.4469	0.04	0.37
Time	0.03643	6	0.006071	$F(6, 138) = 8.76$	<0.0001	0.28	0.99
Maturity	1.878	1	1.878	$F(1, 23) = 191.0$	<0.0001	0.89	1.00
GCT							
Time × maturity	0.0033	6	0.00055	$F(6, 138) = 0.89$	0.7496	0.02	0.22
Time	0.0041	6	0.00068	$F(6, 138) = 1.1$	0.2073	0.06	0.57
Maturity	0.0014	1	0.0014	$F(1, 23) = 0.28$	0.5296	0.02	0.09
RSI							
Time × maturity	0.17	6	0.029	$F(6, 138) = 1.8$	0.089	0.08	0.68
Time	0.79	6	0.13	$F(6, 138) = 8.0$	<0.0001	0.26	0.99
Maturity	31	1	31	$F(1, 23) = 109$	<0.0001	0.81	1.00
SJ height							
Time × maturity	0.001718	6	0.0002864	$F(6, 138) = 1.10$	0.3635	0.05	0.42
Time	0.01585	6	0.002642	$F(6, 138) = 10.18$	<0.0001	0.31	0.99
Maturity	1.154	1	1.154	$F(1, 23) = 118.8$	<0.0001	0.84	1.00

*ANOVA = analysis of variance; SJ = squat jump; DJ₃₀ = drop jump from 30 cm; GCT = ground contact time; RSI = reactive strength index; SS = sum of squares; DF = degree of freedom; MS = mean squares; F = F ratio; η^2 = partial eta-squared.

difference is related to an increase in the muscle fiber size (hypertrophy) caused by a hormonal change (accumulation of androgen hormones and growth hormone) and a change in muscle typology, with an increased percentage of fast twitch fibers from childhood into late adolescence (20). In trained young athletes, strength can be also further enhanced by resistance training (6). The cumulative effect of maturation and training can also translate into an increase in tendon stiffness that facilitates the SSC efficiency (19,36), allowing for better utilization of elastic energy during plyometric type exercises (such as the DJ₃₀). Despite this, the greater force production capacity in postpubertal boys because of growth and maturation as well as the length of exercise training exposure, no maturity-related differences were observed in the time over which vertical jump performance improvements were elicited after the standardized dynamic warm-up. These results are not consistent with a priori expectation because we hypothesized that postpubertal boys would benefit more from the warm-up than prepubertal boys because of their higher percentage of fast twitch fibers (20). Indeed, if postpubertal boys recruit more extensively a greater proportion of fast twitch fibers during maximal exercise, they could generate more metabolic by-products (especially lactate) to drive water into the muscle fibers and hence benefit more favorably from the postactivation performance enhancement (PAPE) effects during high-speed movement, such as a vertical jump. However, the results of this study did not show significant recovery time × maturity group

interaction effects on any parameter measured during vertical jumps, suggesting that the PAPE effect after a dynamic warm-up is not dependent on the maturity status of boys and that the muscle typology might not be necessarily different between prepubertal and postpubertal boys, as previously shown (7). Therefore, the possibility exists that other factors such as intermuscular coordination (e.g., agonist-antagonist coactivation), intramuscular coordination (i.e., recruitment and synchronization of motor units, motor unit firing frequency), and training experience are more important for performance than muscle contractile capacity in complex activities such as vertical jumping. This possibility might be explicitly tested in future studies.

Although no significant recovery time × maturity group interaction effects were observed, the subjects improved their performance in the first minutes after the warm-up, regardless of their maturity status. Specifically, after completion of the warm-up, subjects reached their peak SJ height at 3 minutes and DJ₃₀ height and RSI at 1.5 minutes; RSI was increased directly by an increase in DJ₃₀ height because GCT remained unchanged over time. This suggests that the standardized dynamic warm-up had beneficial effects both on muscle concentric performance and explosive stretch-shorten cycle capacity during vertical jumps but with a delayed optimum effect for the latter. The results of this study are difficult to compare with the existing literature because we are the first to compare changes in vertical jump performance after a standardized dynamic warm-up between children of

Table 5
Vertical jump outcomes obtained before warm-up and across the different post-warm-up time points (mean ± SD).*

	PRE	0 s	1.5 min	3 min	5 min	7 min	10 min
Prepubertal boys							
SJ height (m)	0.166 ± 0.026	0.187 ± 0.035	0.180 ± 0.031	0.189 ± 0.033	0.171 ± 0.029	0.177 ± 0.03	0.172 ± 0.034
DJ ₃₀ height (m)	0.171 ± 0.030	0.194 ± 0.033	0.205 ± 0.042	0.189 ± 0.028	0.196 ± 0.029	0.175 ± 0.037	0.175 ± 0.035
GCT (s)	0.236 ± 0.028	0.226 ± 0.044	0.245 ± 0.052	0.221 ± 0.041	0.230 ± 0.039	0.227 ± 0.054	0.244 ± 0.042
Postpubertal boys							
SJ height (m)	0.326 ± 0.047†	0.348 ± 0.053†	0.354 ± 0.042†	0.357 ± 0.054†	0.335 ± 0.052†	0.332 ± 0.042†	0.328 ± 0.044†
DJ ₃₀ height (m)	0.369 ± 0.046†	0.413 ± 0.053†	0.418 ± 0.049†	0.405 ± 0.052†	0.393 ± 0.056†	0.369 ± 0.073†	0.389 ± 0.046†
GCT (s)	0.241 ± 0.01	0.235 ± 0.016	0.231 ± 0.032	0.236 ± 0.024	0.245 ± 0.019	0.227 ± 0.031	0.244 ± 0.021

*PRE = pre-warm-up performances; SJ = squat jump; DJ₃₀ = drop jump from 30 cm; GCT = ground contact time.

†Significantly different between prepubertal and postpubertal boys at $p < 0.001$.

Table 6**Bonferroni's multi comparisons test for time effect.***

	PRE vs. 0 s	PRE vs. 1.5 min	PRE vs. 3 min	PRE vs. 5 min	PRE vs. 7 min	PRE vs. 10 min
Mean differences						
SJ height (m)	−0.021†	−0.021†	−0.027†	−0.008	−0.008	−0.004
	95% CI = −0.032	95% CI = −0.027	95% CI = −0.018	95% CI = −0.019	95% CI = −0.019	95% CI = −0.014
	to −0.011	to −0.014	to −0.003	to −0.002	to −0.002	to −0.006
DJ ₃₀ height (m)	−0.033†	−0.040†	−0.026‡	−0.023†	−0.002	−0.011
	95% CI = −0.048	95% CI = −0.059	95% CI = −0.045	95% CI = −0.036	95% CI = −0.025	95% CI = −0.024
	to −0.018	to −0.021	to −0.007	to −0.011	to 0.021	to 0.001
RSI	−0.177†	−0.182†	−0.165†	−0.1‡	−0.093	−0.036
	95% CI = −0.241	95% CI = −0.295	95% CI = −0.262	95% CI = −0.168	95% CI = −0.193	95% CI = −0.107
	to −0.112	to −0.07	to −0.07	to −0.03	to 0.007	to 0.03

*PRE = pre-warm-up performances; SJ = squat jump; DJ₃₀ = drop jump from 30 cm; RSI = reactive strength index; CI = confidence interval.†Significantly different at $p < 0.001$.‡Significantly different at $p < 0.01$.

different maturity levels depending on post-warm-up recovery time. However, the results are partly consistent with those of Faigenbaum et al. (16), who showed an increased CMJ height following dynamic exercises as compared with static stretch exercises up to 6 minutes after the completion of the intervention in 16.5-year-old male adolescents.

The 12.0% improvement in SJ height up to 3 minutes after warm-up could be explained by the increase in muscle temperature (8), which seems to be a main cause for the increase in short duration performance (11) because it allows for an increase in

muscle power (9,31). The improvement in SJ performance in the first 3 minutes might also be ascribed to an increase in muscle or cellular water content, which affects the contractile response (i.e., actin-myosin cross-bridge cycle) and may have a similar time course to increases in muscle temperature (12). Alternatively, the improvement in SJ height could be explained by alterations in muscle activation, including a greater spinal excitability (i.e., an increase in the net output of motor neurons) (25) or intermuscular coordination (i.e., technique).

Contrary to our hypothesis, the improvement in SJ height was not significantly greater than DJ₃₀ height (12.0 vs. 16.9%, $p > 0.05$). This finding is not consistent with a priori expectation because we hypothesized that the improvement in SJ height would be greater than DJ₃₀ height because of the complexity of performing a DJ, which requires the use of a stretch-shortening cycle and speculatively a more movement pattern-specific warm-up (1). According to our data, DJ₃₀ performance may have been enhanced through the same mechanisms as SJ, but other processes that allowed for an increase in SSC are also possible. It is conceivable that repeating DJ₃₀ before the warm-up (pretest) and at the end of the warm-up had an impact on movement technique that helped to condition the motor pattern (28). Also, it is noteworthy that the 4 jumps ($2 \times$ SJ and $2 \times$ DJ₃₀) performed at the end of the warm-up did not appear to generate muscle fatigue because no loss of performance was observed immediately after warm-up (i.e., at the 0 minutes time point). The 4 jumps were also specific to performance because they were similar to the tests done after the warm-up. This specificity could have limited motor pattern interference (i.e., motor pattern perseveration that negatively impacts a new motor task when preceded by another motor task), which could have limited the improvement in jump performance (12), notably for the DJ₃₀.

In each session, subjects jumped on a single delay after warm-up (0, 1.5, 3, 5, 7, or 10 minutes) in a randomized order, and intertrial test-retest reliabilities for pre-warm-up outcomes were calculated over the 6 experimental sessions. To ensure an acceptable level of reproducibility, the 6 experimental sessions on different days were completed under the same conditions (same test location, partners, time of day, indoor temperature, and pretest physical activity level). However, the 2 familiarization sessions that were provided to reduce learning effects on SJ and DJ₃₀ performances appear to have been insufficient because reproducibility was slightly lower in prepubertal than in postpubertal boys. The moderate reproducibility reported in prepubertal boys could be explained by the immaturity of central

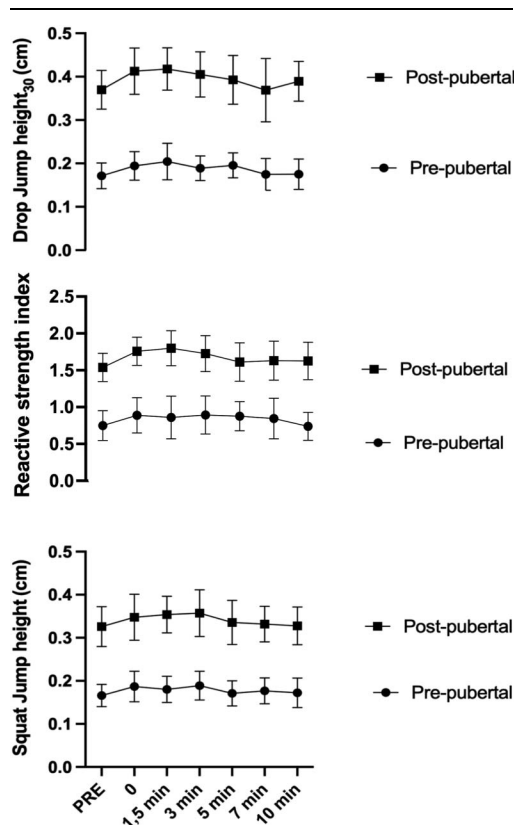


Figure 1. Time course of drop jump height from 30 cm, reactive strength index, and squat jump height across the different post-warm-up time points (0, 1.5, 3, 5, 7 and 10 minutes) in the 2 maturity groups (prepubertal and postpubertal children). PRE = before the warm-up.

nervous system before puberty, reducing motor pattern consistently, notably during complex high-speed polyarticular movements, such as vertical jump (19). However, it can be assumed that the jumps performed during the 2 introductory sessions helped to increase the level of reproducibility over the 6 experimental sessions, whatever the maturity group.

One possible limitation of this study is that training background differed between prepubertal and postpubertal boys at the time of study, and postpubertal children were engaged in a specific sport (i.e., rugby) and strength conditioning sessions contrary to prepubertal boys. However, this reflects real-world training conditions because most prepubertal boys at the age of 9–10 years usually complete fewer hours of deliberate sporting activity per week than their older counterparts, and they do not specialize in a specific sport before the age of 13–14 years (21). Nonetheless, individuals in both cohorts performed various activities as part of their school physical education as well as in daily unstructured play. In addition, the athletes' strength training backgrounds and strength levels should not represent important variables in PAPE because (a) some previous studies showed no differences in PAPE after a conditioning activity between power track and field athletes (training for power), bodybuilders (training for hypertrophy), and physically active (no strength training experience) adult subjects (5), (b) meta-analysis revealed no clear effects (nonoverlap in 95% confidence intervals) between cohorts with different training backgrounds or strength levels (32), and (c) significant correlations were not observed between strength and jumping performance before and after a conditioning activity in strength-trained adult individuals (22). Another possible limitation is that muscle temperature was not measured in this study, yet it is considered as one of the main factors accounting for power output (8), notably in fast-twitch fibers at high muscle contraction velocities (17).

To conclude, the results of this study show that an 8-minute, standardized, dynamic warm-up, including low-intensity, non-specific exercises progressing to maximal specific exercises, as commonly used in sport settings, was beneficial for improving vertical jump performance in children and adolescents. However, the magnitude of performance benefit was not dependent on children's maturity status.

Practical Applications

The standardized dynamic warm-up used in this study had a positive impact on muscle concentric power and stretch-shortening cycle function in young male athletes. Our results suggest that there could be a "window of opportunity" to improve vertical jump performance immediately after the warm-up (performance peaking at 3 minutes for the SJ and at 1.5 minutes for the DJ₃₀), but this window does not appear to be dependent on the maturity status of children. This performance improvement can be applied in short duration disciplines, such as some jumping disciplines in athletics with significant recovery times between the actions. From a practical perspective, it will be important to adapt the warm-up to the constraints of the competition. However, practitioners should keep in mind that many factors, including the highly individualized nature of the PAP/PAPE response, ambient temperature, fatigue, recovery position, and mode (sitting, standing, walking), and warm-up protocol may influence the optimal recovery period vertical jump performance improvement after the warm-up.

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